

Patterns of Truth: Location and Modality

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Introduction

Truth Patterns: Time

Truth varies across times in tense logic.

- (1) There is a total lunar eclipse.
- (2) Humans first walk on the Moon.

Both propositions are false now, but they have been true at different time intervals. One has most recently been true on March 3, whereas the other has been true on July 21, 1969.

They differ in *temporal profile*. That is, they each have a distinctive pattern of truth across *time intervals*.

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Truth Patterns: Space

If truth varies across places, then propositions have a *spatial profile*.

(3) The Moon is visible.

(4) It is daytime.

One proposition is now false in Barcelona, but true in Auckland.

They differ in *spatial profile*. That is, they each have a distinctive pattern of truth across *places*.

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Truth Patterns: Modality

If truth varies across worlds, then propositions have a *modal profile*.

(5) There are human outposts on the Moon.

(6) There is a US presidential election in 2026.

Both propositions are false in the actual world, but they may be true at different possible worlds.

They differ in *modal profile*. That is, they each have a distinctive pattern of truth across *worlds*.

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Locational Profile

We will just speak of **locations** as we develop a general unified framework under which to subsume space, time, and modality.

Complex locations subsume further sublocations as parts:

- a *time interval* is composed of subintervals (and instants).
- a *spatial region* is composed of subregions (and points).
- a *possibility* is composed of finer possibilities (and worlds).

What exactly is for a proposition to be true at a complex location?

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Some distinctions

(2) Humans first walk on the Moon.

- true *all over* a location
(2) is true *all over* multiple one-minute intervals on July 21, 1969. That is, true at all of their subintervals.
- true at *some of* a location
(2) is true at some of July 21, 1969. That is, true at some subintervals of the twenty four-hour interval.
- true *entirely within* a location
(2) is true *entirely within* July 1969. That is, true at some of that interval but not true outside that interval.

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- true *exactly at* a given location
(2) is true *exactly at* an approximately two-hour interval on July 21, 1969.

There has recently been a total lunar eclipse on March 3.

(1) There is a total lunar eclipse.

- Is (1) true *entirely within* the totality interval on March 3?
- Is (1) true *exactly at* the totality interval on March 3?

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Functionality claims that objects have *at most one* exact (temporal) location and seems inconsistent with endurance.

We explore the parallel with propositional truth.

- Is there a non-functional form of *true exactly at*?
- Is a non-functional form *true exactly at* reducible to more basic locational relations such as *true all over*?

We will give different answers to these questions.

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Why Care?

This is not just a formal exercise:

Persistence and Location Some proponents of endurance posit a non-functional relation of exact location, which is not reducible to more basic relations such as weak, pervasive, and entire location.

- Socrates exists at noon and at 1pm.
- Socrates is *exactly* (temporally) *located* at noon and *exactly* (temporally) *located* at 1pm.
- Socrates (temporally) *pervades* noon and (temporally) *pervades* 1pm.
- Socrates is neither *entirely* (temporally) *located* at noon nor *entirely* (temporally) *located* at 1pm.

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The Framework

Roadmap

We explain these distinctions in a modal language $\mathcal{L}_{\Box, \Diamond}$

- equipped with two basic modal operators $\Box$ and $\Diamond$, and
- interpreted over a system of locations M ordered by the relation of whole to part.

We will use a modality to capture a locative relation between propositions and locations. For example:

Locative Relation	Modality	Definition
φ is true all over u in M	u verifies $@_{>}\varphi$ in M	$M, u \Vdash \Box\varphi$
φ is true at some of u in M	u verifies $@_{\circ}\varphi$ in M	$M, u \Vdash \Diamond\varphi$

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Locational Frames

There are other targets besides pervasive and weak location:

Locative Relation	Modality	Definition
φ is true entirely within u in M	u verifies $@_{<}\varphi$ in M	TBA
φ is perfectly true at u in M	u verifies $@_{=}\varphi$ in M	TBA
φ is true exactly at u in M	u verifies $\text{Ex}\varphi$ in M	None

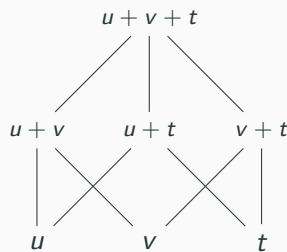
We interpret the language over locational frames $(L, \sqsubseteq)$, where:

- L is a set of locations
- $\sqsubseteq$ is the relation of **whole to part**.

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Locational Frames

$(L, \sqsubseteq)$ is a *partial order*: reflexive, anti-symmetric, transitive.
Maybe even a model of Classical Mereology (CM):



Example

Let u , v , and t one-hour intervals in July 19, 1969. $u + v$ is the sum of the first two intervals; $u + v + t$ is the sum of all three; etc. The first Moon walk is entirely located within $u + v + t$.

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Extended Modal Language

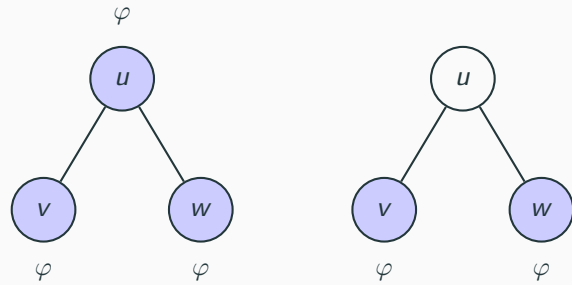
We expand $\mathcal{L}_{\Box}$ with a *window* modality: $\Box\Box$.

$M, u \Vdash \Box\varphi$ iff for all $v \in L$, if $u \sqsubseteq v$, then $M, v \Vdash \varphi$.
all sublocations of u verify φ

$M, u \Vdash \Box\Box\varphi$ iff for all $v \in L$, if $M, v \Vdash \varphi$, then $u \sqsubseteq v$.
only sublocations of u verify φ

If $u \Vdash \Box\Box\varphi$, then u acts like a *window* from which all φ -locations are accessible.

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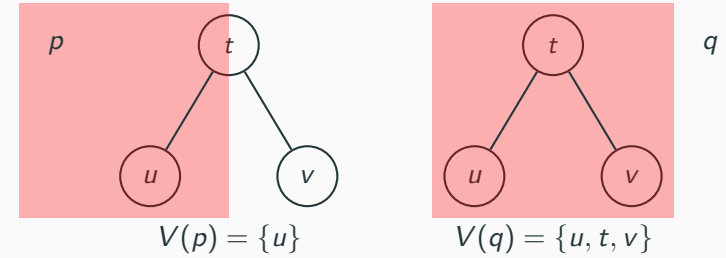


$u \Vdash \Box \varphi$ all sublocations of u verify φ
 $u \Vdash \Box \Box \varphi$ only sublocations of u verify φ

If φ is true entirely within u , u includes all φ -locations as parts.

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Atomic Verification



We gloss $t \in V(p)$ as the claim that p is true *all over* t .

Assignment

An **assignment** V for a locational frame $(L, \sqsubseteq)$ is a map $V : \text{AT} \rightarrow \mathcal{P}(L)$ subject to a **heredity constraint**:

- if $t \sqsubseteq v$, then if $t \in V(p)$, then $v \in V(p)$.

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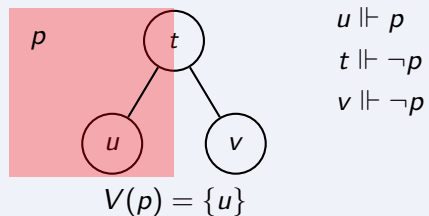
Verification

Definition

If M is a model $(L, \sqsubseteq, V)$, then for each $u \in L$:

$M, u \Vdash p$ iff $M, u \in V(p)$
 $M, u \Vdash \neg \varphi$ iff $M, u \nVdash \varphi$
 $M, u \Vdash (\varphi \wedge \psi)$ iff $M, u \Vdash \varphi$ and $M, u \Vdash \psi$

Example



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Locational Modalities

Locational Modalities

pervasive truth

$$@_{>} \varphi := \Box \varphi$$

all sublocations verify φ

true all over

weak truth

$$@_{\circ} \varphi := \Diamond \varphi$$

some sublocation verifies φ

true at some of

entire truth

$$@_{<} \varphi := \Diamond \varphi \wedge \Box \varphi$$

only sublocations verify φ but some do

true entirely within

perfect truth

$$@_{=} \varphi := \Box \varphi \wedge \Box \Box \varphi$$

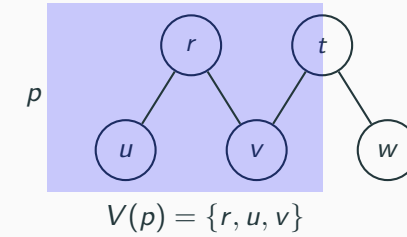
all and only sublocations verify φ

$$\blacksquare \varphi$$

true all over and entirely within

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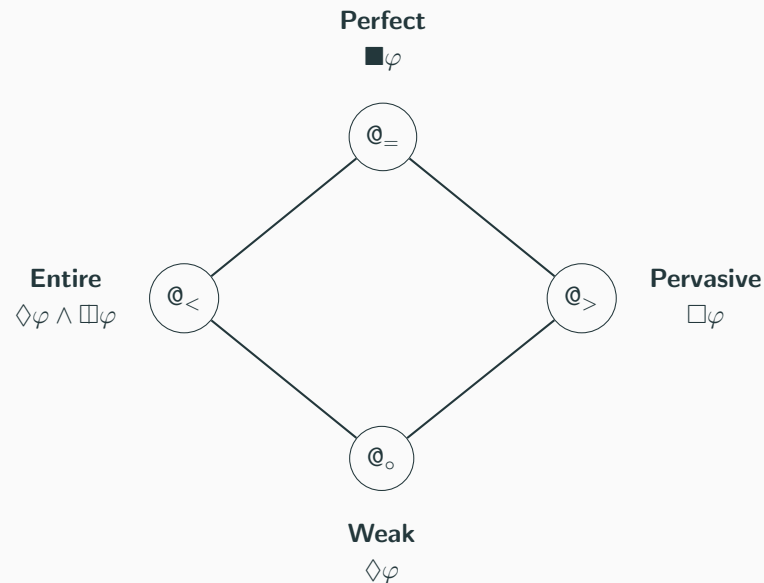
Example



Modality	Example	Justification
Weak	$t \Vdash @_{\circ} p$	$t \sqsupseteq v$ and $v \Vdash p$
Pervasive	$u \Vdash @_{>} p$	$x \Vdash p$ if $u \sqsupseteq x$
Entire	$r \Vdash @_{<} p$	$x \Vdash p$ only if $r \sqsupseteq x$
Perfect	$r \Vdash @_{=} p$	$x \Vdash p$ iff $r \sqsupseteq x$

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Entailment



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Exact Location

(1) There is a total lunar eclipse

- (1) is **true exactly at** an approximately one-hour interval on March 3.
- (1) is **not true entirely within** a one-hour interval on March 3.
- (1) is **not perfectly true at** an approximately one-hour interval on March 3.

Moral

To be **perfectly true at** a location is not the same as to be true **exactly at** that location.

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Perfect Location is Functional

If a proposition is perfectly true at some location, then it is perfectly true nowhere else.

Remark

All **locational** frames are functional.

This is a simple consequence of the anti-symmetry of $\sqsubseteq$.

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Exact Location is not Functional

Some propositions are true at exactly more than one location.

(1) There is a total lunar eclipse.

- (1) is true exactly at the totality interval on March 3.
- (1) is true exactly at the totality interval on September 7.

If (1) is true exactly at a time interval I , then I is a perfect location for a *truthmaker*, which is itself not a proposition.

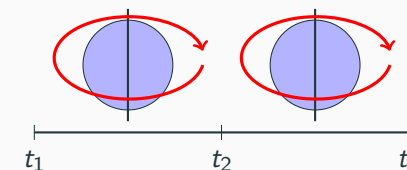
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Exact Location is not Additive

Even if a proposition is true exactly at u and true exactly at v , the proposition may fail to be true exactly at $u + v$.

(7) The sphere makes one complete rotation.

There are two *truthmakers* depicted in the diagram:



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What to Make of Exact Location?

1. How should we characterize exact location?

Enrich $\mathcal{L}_{\square, \Box}$ with a new modality Ex for exact location. We explain how to interpret it over a richer class of models.

2. Is exact location definable within $\mathcal{L}_{\square, \Box}$?

Undefinability Thesis

Exact location is not definable within $\mathcal{L}_{\square, \Box}$.

Exact location is not just a matter of what complex pattern of truth across locations a proposition exemplifies.

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The Realization Framework

- (1) There is a total lunar eclipse.

- The totality interval on March 3 **realizes** (1).
- The totality interval on September 8, 2025 **realizes** (1).

Do locations ever realize complex propositions?

- (8) It is not the case that there is a total lunar eclipse.

- (9) There is a total lunar eclipse and there is a US election.

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Some constraints on realization:

- u *realizes* φ only if u *pervades* φ .
- if φ is *entirely located* at u , only sublocations of u *realize* φ .

If φ is *perfectly located* at u and V is a set of *realizers* for φ :

- for all $v \in V$, $u \sqsupseteq v$.
- if there is such a location as $\bigvee V$, then $\bigvee V \sqsupseteq u$.

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Tentative Proposal

Negation

If φ has a perfect location u , then if the complement $\bar{u}$ exists, $\bar{u}$ is a realizer for $\neg\varphi$. Otherwise, $\neg\varphi$ *lacks a realizer*.

The complement of the perfect location of (1) realizes the negation (8) above.

Conjunction

If φ and ψ have each a set of realizers U and V , then a location t realizes the conjunction iff t realizes both conjuncts φ and ψ .

Some time interval in Nov 8 2022 realizes the conjunction (9) above because it realized each conjunct.

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Realization Framework: Formal Details

Realization Assignment

A **realization assignment** for M is a map: $r : \text{At} \rightarrow \mathcal{P}(L)$, where:

$$\begin{aligned} r(p) &\subseteq \llbracket \Box p \rrbracket_M && \text{realizations pervade} \\ u \in \llbracket \Box(p) \rrbracket_M &\text{ iff } \forall v \in r(p) \ u \sqsupseteq v && \text{realizations are bounded} \end{aligned}$$

Each assignment determines a realization function:

Realization Function

$$\begin{aligned} R(p) &= r(p) \\ R(\neg\varphi) &= \{\Box R(\varphi)\} \text{ if it exists, else } \emptyset \\ R(\varphi \wedge \psi) &= R(\varphi) \cap R(\psi) \end{aligned}$$

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Exact Location

We operate in an expanded language $\mathcal{L}_{\Box, \Box, \text{Ex}}$, where:

$$M, u \Vdash \text{Ex } \varphi \text{ iff } u \in R(\varphi)$$

That is, u verifies $\text{Ex } \varphi$ iff u is a **realization** for φ .

We may now define **weak entire location**:

$$\begin{aligned} M, u \Vdash @_{\prec} \varphi &\text{ iff } M, u \Vdash \neg \Box \neg \text{Ex } \varphi \\ &\text{ } u \text{ is part of some exact location for } \varphi \end{aligned}$$

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Undefinability

Remark

The exact location operator Ex is not definable in $\mathcal{L}_{\Box, \Box}$.

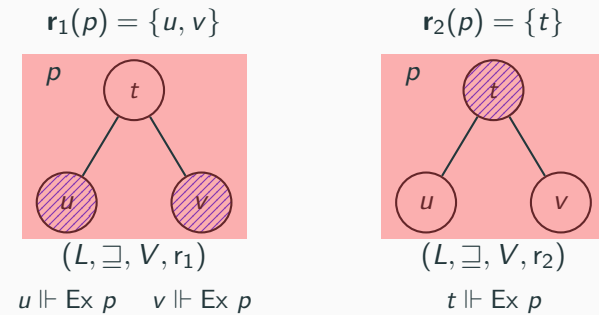
Expand a model M for $\mathcal{L}_{\Box, \Box}$ into models M_1 and M_2 for the expanded language $\mathcal{L}_{\Box, \Box, \text{Ex}}$, which

- agree with respect to formulas in $\mathcal{L}_{\Box, \Box}$, yet
- disagree with respect to formulas of the form $\text{Ex } \varphi$.

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Models

We expand $(L, \sqsupseteq, V)$ with two realization assignments r_1 and r_2 :

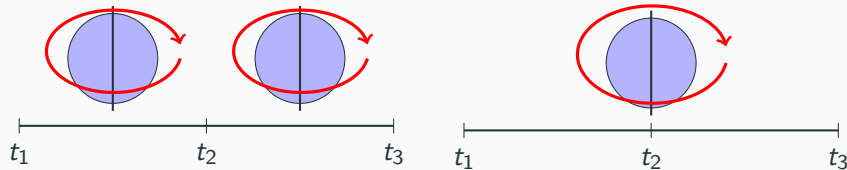


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The Rotation of the Sphere Revisited

To make the model vivid, consider:

(3) The sphere completes one rotation.



(3) is exactly true at (t_1, t_2) **and**
 (t_2, t_3) but not at (t_1, t_3) .

(3) is exactly true at (t_1, t_3) but
not at (t_1, t_2) or (t_2, t_3)

$\mathcal{L}_{\square, \boxplus}$ remains blind to the distinction.

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Conclusion

Conclusion

We have introduced an extended modal framework, which

- supplements $\mathcal{L}_{\square}$ with a window operator $\boxplus$
- interprets the language over frames $(L, \sqsubseteq)$, which encode the mereological structure of a system of locations

We used the extended modal framework to:

1. characterize a family of locational modalities, e.g., pervasive, entire, and perfect location.

We asked whether exact location is in principle definable in terms of the distribution of truth values across locations.

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In order to answer that question, we

2. introduced a new locational modality for exact location
3. argued that exact location is undefinable in $\mathcal{L}_{\square, \boxplus}$

The moral of the discussion is that exact location is not just a pattern of truth across locations.

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